pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Chunking: Size and Overlap

WEEK 1 · DAY 3

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Today: chunking, size and overlap

Two settings, one 10-K

512/64	Jeff's project's setting
1,024/128	the larger alternative
Pieces	136 from day 2
Tokenizers	MiniLM, bge-small, gte-small
Model	none today; nothing is sent

Today

1. Why chunk; Course 7's chunker
2. recursive_split, docstring first
3. chunk_table: the whole 10-K
4. What the overlap carries
5. Where the cuts fall; a debt table
6. Word pieces against the limit
7. Metadata and the running header

Measure what the splitter did before anything is embedded.

Why chunk, and Course 7's chunker

Three limits

A model's context: a 10-K does not fit in a useful request; three short passages do.

A retriever's unit: a chunk about one topic scores high on a question about that topic.

A citation's unit: each chunk carries its item and page, for a reader to check.

421,643
characters in 24 items

0.36%
of it in three chunks of 512

444
chunk_text chunks, ~1,134, chars

0
line breaks left in them

Course 7's word chunker joins words with spaces: a table loses its row ends.

How the splitter works

The problem	An item is far longer than a chunk; a blind cut every 512 characters lands inside words, rows, sentences
The seams	Cut on the first separator; a piece still too long is cut on the next, down to single characters
The merge	Pieces joined back in order while they fit; each new chunk opens with the last overlap characters
You control	The size in characters, the overlap, and the separator list
What comes out	Chunks of at most size characters that rejoin to the input exactly once the overlaps are dropped

A recursive splitter, docstring first

```
SEPARATORS = ("\n\n", "\n", ". ", " ", "")    # paragraph, line, sentence, word, anywhere
```

1. Separators	in order of preference; "" means anywhere, even inside a word
2. Recursion	a piece still longer than size - overlap is cut on the next separator
3. Merge	pieces joined back in order while the chunk fits in size
4. Overlap	each new chunk starts with the last overlap characters of the one before

Tested: drop each chunk's overlap, join, and the input comes back exactly.

By hand: a small debt table at 160/24

Page 37, "Indebtedness": 354 characters, 3 chunks (quoted as published)

chunk 0, 152 characters:

```
December 31, December 31,\n(In millions) 2025 2024 Decrease\nCredit facility $ 160.7 $ 1,300.3 $ (1,139.6)\n4.750% se...
```

chunk 1, 152 characters:

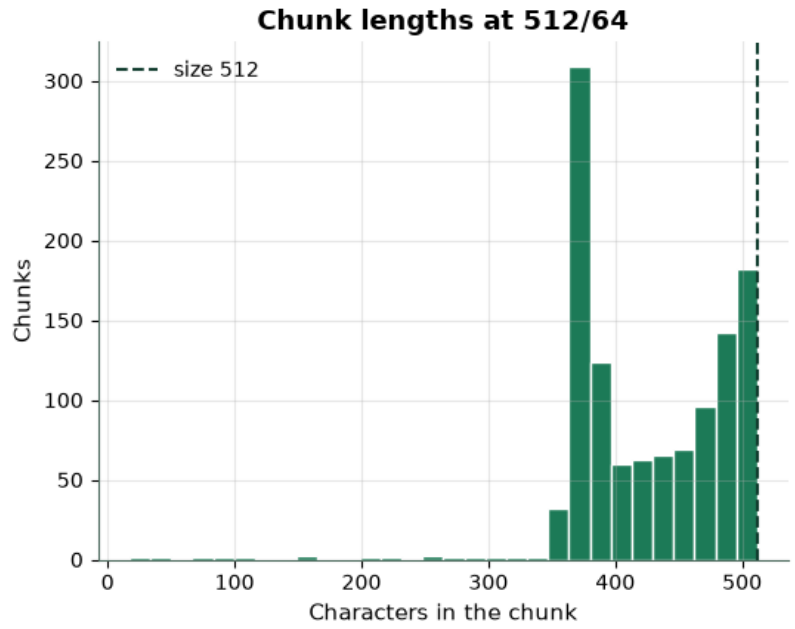
```
 2027 1,000.0 1,000.0 -\n4.750% senior notes due 2031 900.0 900.0 -\nTotal long-term debt 2,060.7 3,200.3 (1,139.6)\n...
```

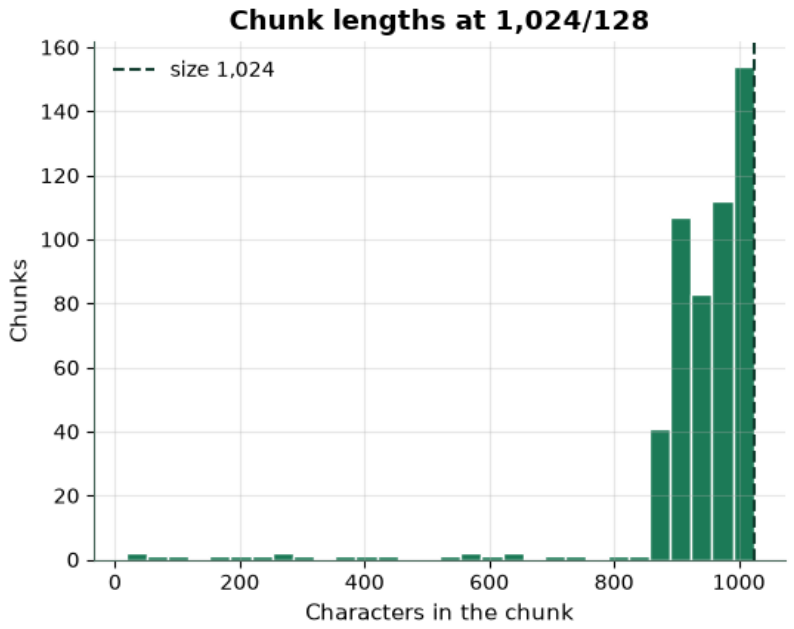
chunk 2, 98 characters:

```
aturities - 44.0 (44.0)\nLong-term debt, net of current maturities $ 2,060.7 $ 3,156.3 $ (1,095.6)\n
```

Every chunk ends on \n, a row's end; the next starts 24 characters back, mid-row.

The whole 10-K at two settings





chunk_table

1,158

chunks at 512/64

520

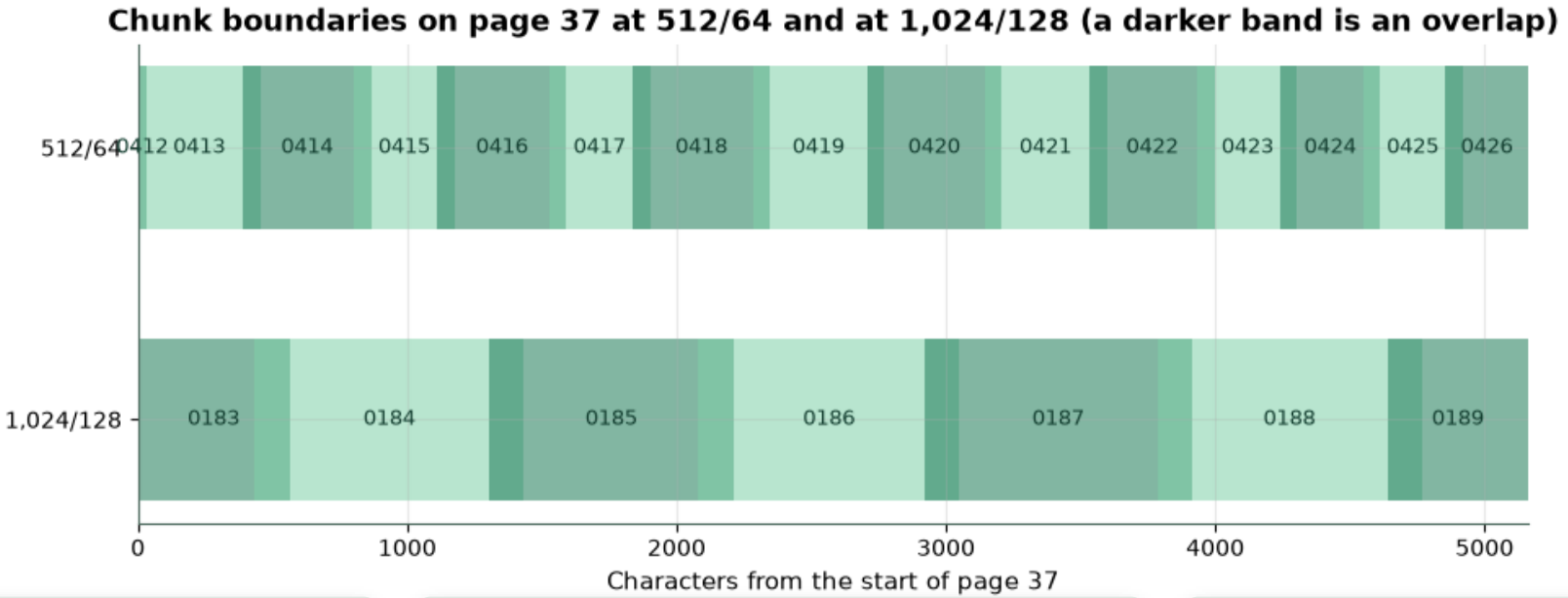
chunks at 1,024/128

~370

the tallest bar at 512/64

A prose line is about 150 characters: 64 carried + two whole lines, and a third does not fit.

One page, its chunk boundaries at two settings



15

chunks touch page 37 at 512/64

7

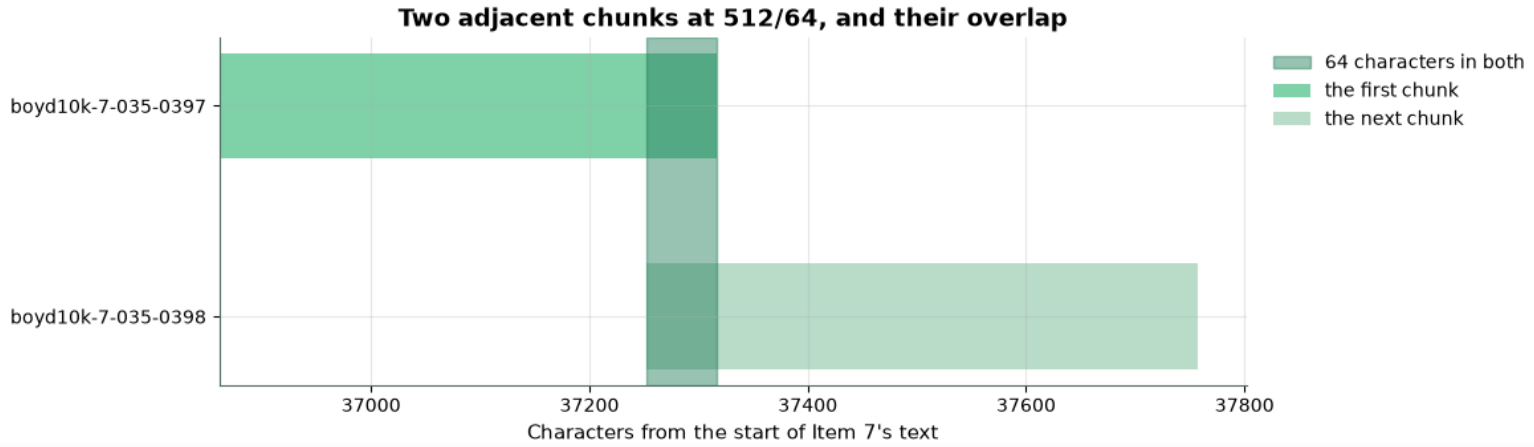
at 1,024/128

darker

the overlap: 64 or 128 characters

The same page, and the cut points move. The first band in each row started on page 36, which its citation names.

What the overlap carries



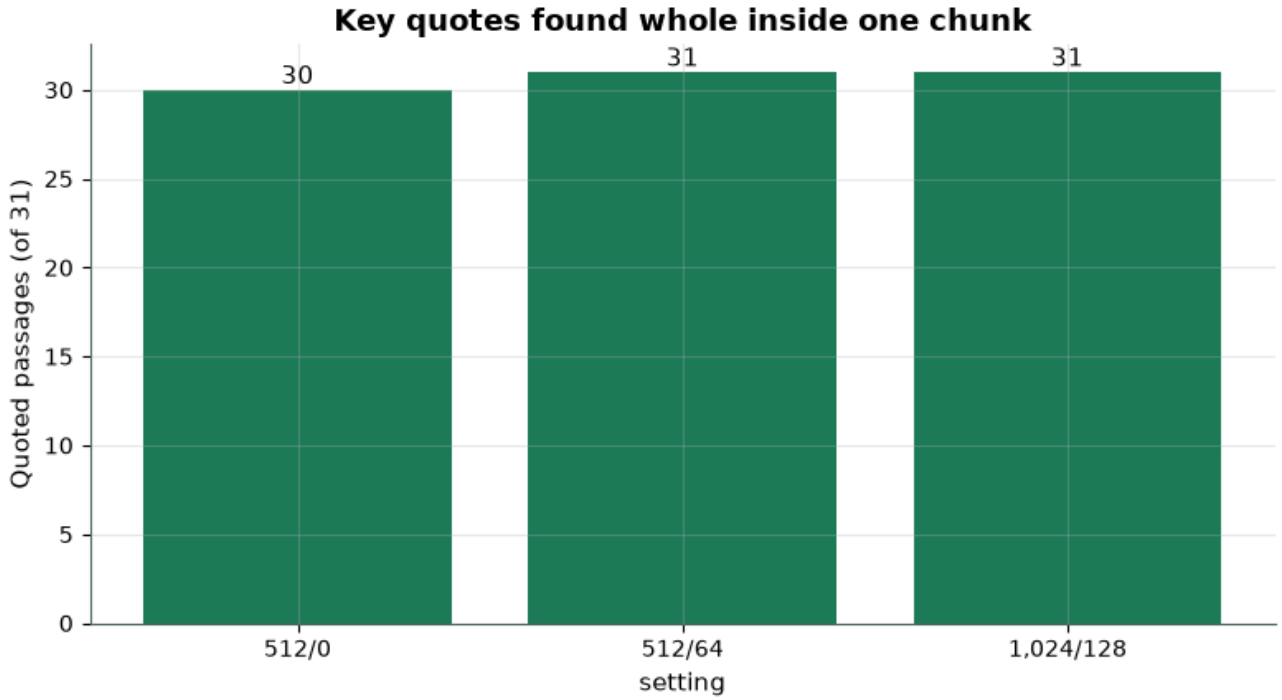
The next chunk's first lines (quoted as published)

```
nimize borrowings and related interest costs. Our cash and cash
equivalents balances were $353.4 million and $316.7 million at December 31, 2025 and 2024, respectively. In...
```

1,134 of 1,134 adjacent pairs share exactly the last 64 characters. It starts mid-word; its cut was on a line break.

Source: Boyd Gaming Corporation, Form 10-K for fiscal 2025, filed 2026-02-20, accession 0001437749-26-004908, Item 7, page 35 of the course's PDF.

Key quotes: whole in one chunk, or cut apart



512/0 cuts dev14 (as published)

```
...ur working capital deficit at December 31,
2025 and 2024 was
```

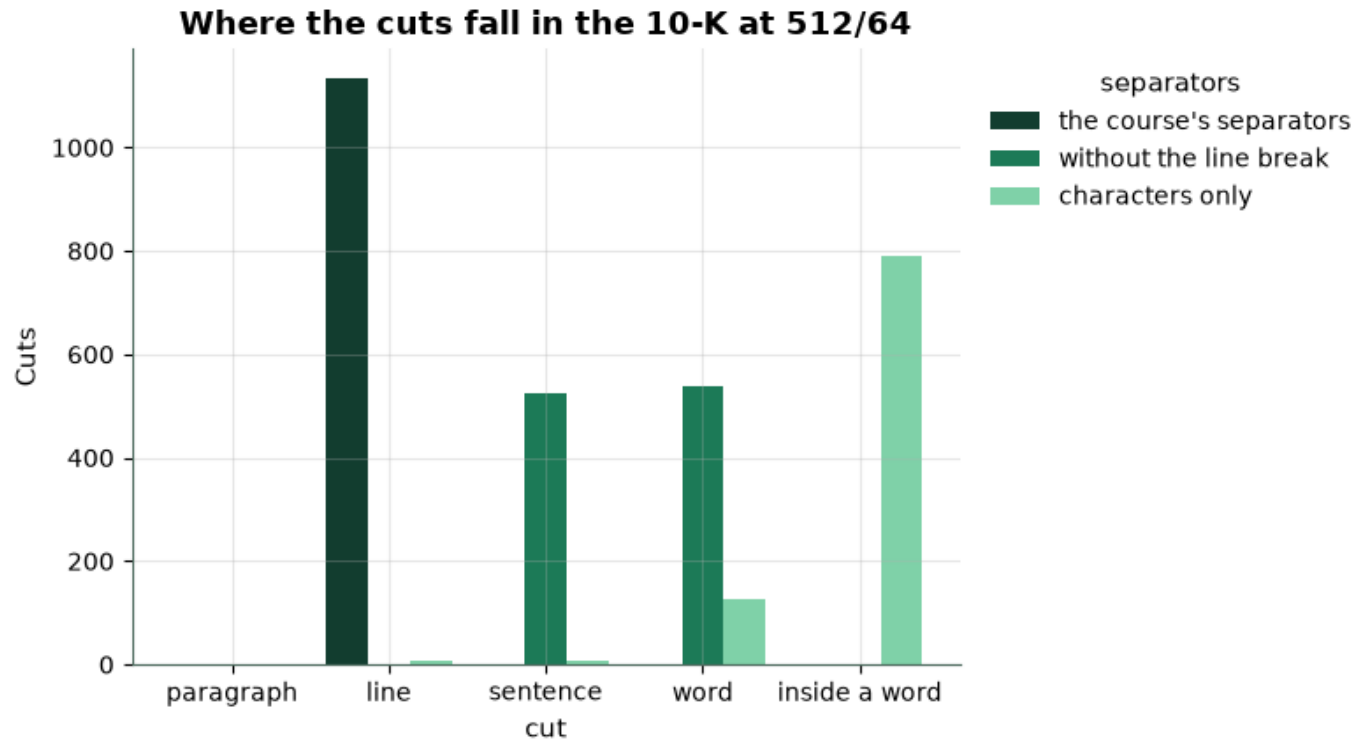
| the next chunk:

```
$448.5 million and $61.2 million, respectively.
The increase in our...
```

The subject in one chunk, its figures in the next.

With 64 overlapping, and at 1,024/128, all 31 quoted passages are whole.

Where the cuts fall



What the census found

1,134

cuts, every one on a line break

0

paragraph breaks in the PDF's text

14.46%

of them after a full stop

characters only

the cuts not on a separator

A line break is a separator, not a sentence end: the printer wrapped those lines.

Note 7's debt table, cut three ways

2025 block, p. 74: the next chunk starts with the carried heads (quoted as published)

```
ates at Origination
December 31, Outstanding Fees and Long-Term
(In thousands) 2025 Principal Costs Debt, Net
```

the course's separators: the cut is line

```
'nior notes due 2031 4.750% 900,000 (8,763) 891,237\nOther 5.208% 6 - 6\n' | 'Total long-term debt 3,200,306 (23,71...
```

without the line break: the cut is word

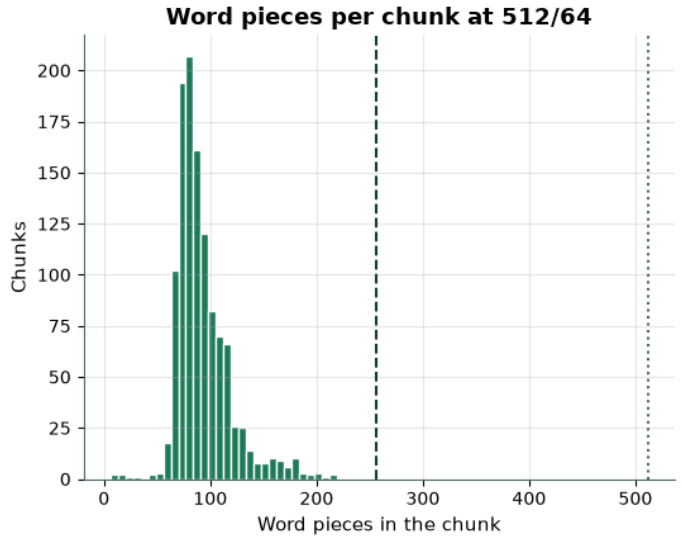
```
'rm debt 3,200,306 (23,716) 3,176,590\nLess current maturities 44,006 - ' | '44,006\nLong-term debt, net $ 3,156,30...
```

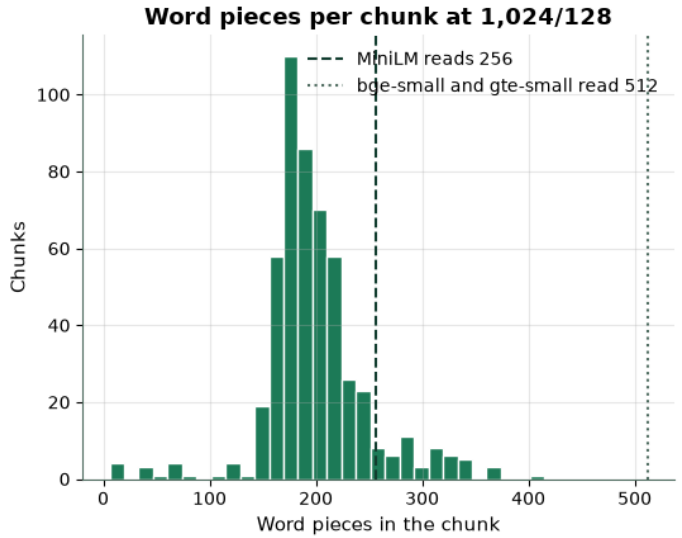
characters only: the cut is inside a word

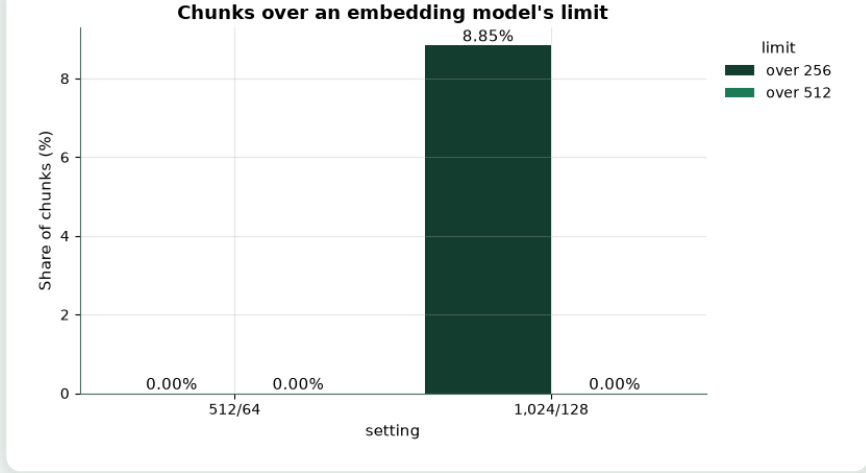
```
'enior notes due 2027 4.750% 1,000,000 (5,844) 994,156\n4.750% senior no' | 'tes due 2031 4.750% 900,000 (8,763) 891,'
```

Source: Boyd Gaming Corporation, Form 10-K for fiscal 2025, filed 2026-02-20, accession 0001437749-26-004908, Item 8, page 75 of the course's PDF.

Word pieces against the model's limit







same

count, three tokenizers

219

longest at 512/64

46

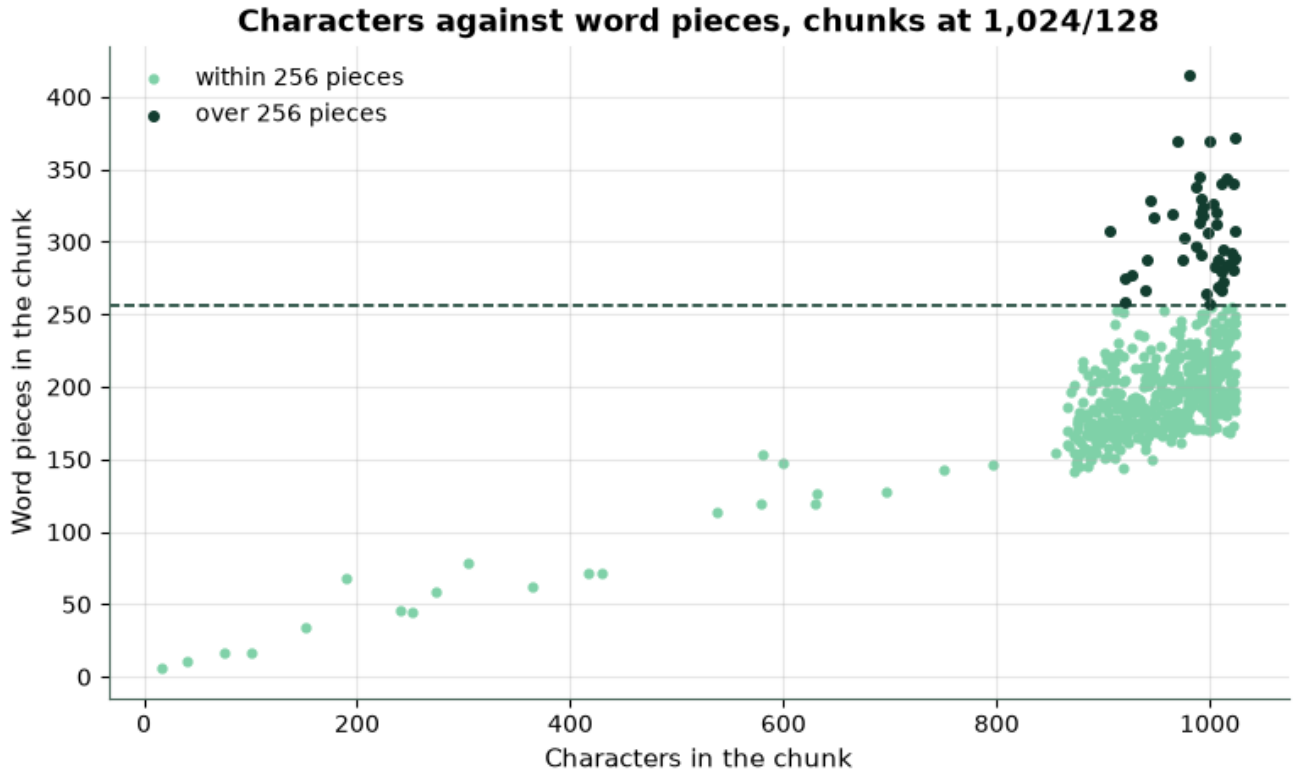
over 256 at 1,024/128

0

over 512, both settings

MiniLM reads 256 pieces; bge-small and gte-small read 512. Past the limit: dropped, no message.

Where the long chunks are, and what MiniLM never reads



In the 46 chunks over 256

7,653

of 45,503 characters never read

16.82%

of their text, with no message

4.70

characters a piece over all chunks

415 pieces

boyd10k-8-072-0354, 980 characters

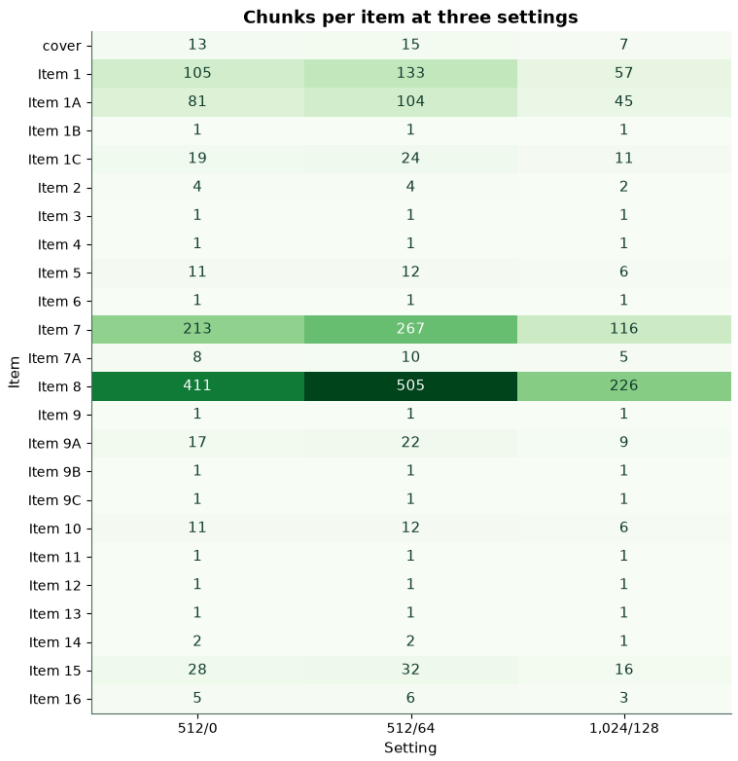
Choose the chunk size in the word pieces of the model you will use.

Every setting of the splitter

10 settings read from the signatures, and 3 model limits from the pins cell

recursive_split	text	required	the text to cut: one item's pieces joined in page order
recursive_split	**size**	512	course: 512 (1,024 compared)
recursive_split	**overlap**	64	course: 64 (128 at 1,024)
recursive_split	**separators**	('\n\n', '\n', '. ', ' ',...	course: SEPARATORS
chunk_table	pages_with_items	required	course: page_items(pages, items)
chunk_table	size	required	course: 512
chunk_table	overlap	required	course: 64
chunk_table	doc	required	course: boyd10k
piece_lengths	tokenizer	required	course: MiniLM's (the three agree here)
piece_lengths	texts	required	course: the chunk table's text column
the model's limit	minilm	256	the most pieces the model reads; a longer chunk is cut ther...
the model's limit	bge_small	512	the most pieces the model reads; a longer chunk is cut ther...
the model's limit	gte_small	512	the most pieces the model reads; a longer chunk is cut ther...

Chunks per item, and the running header



"Table of Contents": kept

512/64: 136 of 1158 hold it, 0 only it; stripped 1156

1,024/128: 118 of 520 hold it, 0 only it; stripped 517

3 pieces each; 9 header-only pieces join the item before.

Same SHA-256 as the committed embeddings at both settings: yes, yes. Day 4 loads them; strip before the first embedding, never after.

What chunking cannot fix

Overlap costs chunks: 1,158 at 512/64, 938 at 512/0. A fact in Items 7 and 8 needs two chunks: retrieval's job.

Excel side by side, and Git

Excel side by side (tested in Excel)

`=LEN(B2), Text cells`	equal on 16 of 16 chunks
`=RIGHT(B2,64)=LEFT(B3,64)`	TRUE on 15 of 15; 13 of 13 cut by characters
`=RIGHT(B2,64)=LEFT(B3,64), the AI check's splitter`	TRUE on 0 of 7: it catches the bug
`=LEN(B2)-LEN(SUBSTITUTE(B2," ",""))+1`	equal on 0 of 16: a line break is not a space
`SUBSTITUTE(B2,CHAR(10)," ") inside TRIM`	equal on 16 of 16

Git: chunks_512_64.csv and chunker.json, ragkit with 20 tests; the switch is a diff.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the overlap counts lines, not characters, so chunks share nothing.

Code that runs is not the same as code that is right.

What you can do now

- Say why a retriever works on chunks
- Write a recursive splitter from its docstring
- Cut a filing inside its items; read ids and citations
- Show what an overlap carries; find a cut off a separator
- Choose a chunk size inside the model's piece limit

Week 1, Day 4 Embeddings and a vector index